Hunter Mills

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Seasoned Data Scientist and Machine Learning Engineer with 8+ years of end-to-end AI product delivery. Proven ability to translate challenges into scalable, production-grade solutions that integrate with medical systems, cloud platforms, and embedded applications. 15+ publications ranging from rare disease prediction, bias mitigation, to deep learning on EKG.

Experience

- **Senior Machine Learning Engineer** — October 2025 - Present
  Visual Thought Labs — San Francisco, CA
  - Developed ambient drug OCR in VA operating rooms, deployed on Nvidia Jetson at 77% lower hardware cost and 73% lower latency; built the SQL pipeline populating it, and HL7 system issuing results.
- **Senior Data Scientist** — January 2022 - October 2025
  UCSF — San Francisco, CA
  - Built “Drifter,” a data-drift monitoring tool used by 80 researchers across six hospitals; surfaced feature, label, and prediction drift that triggered timely ETL updates.
  - Directed a nation-wide matched-cohort study on traumatic-brain-injury outcomes, quantifying a 67% higher risk of malignant brain tumors (HR = 1.67) and publishing findings in JAMA Network Open.
- **Machine Learning Engineer** — October 2021 - January 2022
  Syntegra — San Francisco, CA
  - Created a GenAI synthetic patient data generator that transformed structured EHR records into natural-language sentences, trained BERT models, and regenerated structured data, enabling safe data sharing for rare-case research.
- **Lead Data Scientist** — December 2020 - October 2021
  LifeBell AI — Atlanta, GA (remote)
  - Deployed early sepsis detection system using XGBoost, with clinical explainer at Naval Medical Center San Diego.
  - Designed a Dynamic-Time-Warped Gaussian Mixture Model for phenotype discovery, surfacing sepsis-risk subgroups several hours before conventional detection.
- **Data Scientist** — August 2018 - December 2020
  UCSF — San Francisco, CA
  - Developed deep-learning EKG classifier for pulmonary hypertension, achieving an AUC of 0.89 and early-diagnosis capability up to two years before conventional testing.
  - Led feasibility analyses that helped secure $5M+ in research partnerships (J&J, Roche, Genentech).
- **Embedded Systems Engineer** — May 2014 - August 2018
  NASA Education & Public Outreach — Rohnert Park, CA
  - Conceived/built/launched EdgeCube 10cm satellite, overseeing sensor integration, data logging, and radio telemetry.
  - Developed a STEM curriculum (“Learning By Making”) that raised high-school science scores by 7.8% versus controls.

Education

Stanford University	MS in Computational and Mathematical Engineering	2018
Sonoma State University	BS in Physics and Mathematics	2014

Skills

Languages	Python, SQL, Julia, C, C++, JavaScript, TypeScript
ML Frameworks	PyTorch, TensorFlow, Scikit-Learn, XGBoost, CVXPY, RAPIDS, JAX, TensorRT, Ollama, HuggingFace, StatsModels, lifelines
Infrastructure & Compute	Spark, Spark-SQL, Hadoop, MPI, OpenMP, CUDA, AWS, GCP, Docker

Publications (Selected)

- C. Halabi, **H. Mills**, A. M. DiGiorgio, et al. “Civilian Traumatic Brain Injury Is Associated with Longitudinally Increased Risk of PTSD, Suicidality, and Other Mental Health Diagnoses”. In: *Journal of Neurotrauma* (2026), p. 08977151251414145.
- A. S. Tang, K. P. Rankin, G. Cerono, S. Miramontes, **H. Mills**, J. Roger, et al. “Leveraging electronic health records and knowledge networks for Alzheimer’s disease prediction and sex-specific biological insights”. In: *Nature Aging* 4.3 (2024), pp. 379–395.
- M. A. Aras, S. Abreau, **H. Mills**, L. Radhakrishnan, et al. “Electrocardiogram detection of pulmonary hypertension using deep learning”. In: *Journal of cardiac failure* 29.7 (2023), pp. 1017–1028.